

附录:

泛函网络的主体程序

```
p=P';
Pr=[min(p(1,:)),max(p(1,:));
    min(p(2,:)),max(p(2,:));
    min(p(3,:)),max(p(3,:));
    min(p(4,:)),max(p(4,:));
    min(p(5,:)),max(p(5,:));
    min(p(6,:)),max(p(6,:));
    min(p(7,:)),max(p(7,:));
    min(p(8,:)),max(p(8,:));
min(p(9,:)),max(p(9,:));
    min(p(10,:)),max(p(10,:));
    min(p(11,:)),max(p(11,:));
    min(p(12,:)),max(p(12,:));
    min(p(13,:)),max(p(13,:));
    min(p(14,:)),max(p(14,:));
    min(p(15,:)),max(p(15,:));
    min(p(16,:)),max(p(16,:));
min(p(17,:)),max(p(17,:));
    min(p(18,:)),max(p(18,:));
    min(p(19,:)),max(p(19,:));
    min(p(20,:)),max(p(20,:));
    min(p(21,:)),max(p(21,:));
    min(p(22,:)),max(p(22,:));
    min(p(23,:)),max(p(23,:));
    min(p(24,:)),max(p(24,:));
min(p(25,:)),max(p(25,:));
    min(p(26,:)),max(p(26,:));
    min(p(27,:)),max(p(27,:));
    min(p(28,:)),max(p(28,:));
    min(p(29,:)),max(p(29,:));
    min(p(30,:)),max(p(30,:));
    min(p(31,:)),max(p(31,:));
    min(p(32,:)),max(p(32,:));
min(p(33,:)),max(p(33,:));
    min(p(34,:)),max(p(34,:));
    min(p(35,:)),max(p(35,:));
    min(p(36,:)),max(p(36,:));
    min(p(37,:)),max(p(37,:));
    min(p(38,:)),max(p(38,:));
    min(p(39,:)),max(p(39,:));
    min(p(40,:)),max(p(40,:));
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min(p(41,:)),max(p(41,:));
    min(p(42,:)),max(p(42,:));
    min(p(43,:)),max(p(43,:));
    min(p(44,:)),max(p(44,:));
    min(p(45,:)),max(p(45,:));
    min(p(46,:)),max(p(46,:));
    min(p(47,:)),max(p(47,:));
    min(p(48,:)),max(p(48,:));
min(p(49,:)),max(p(49,:));
    min(p(50,:)),max(p(50,:));
    min(p(51,:)),max(p(51,:));
    min(p(52,:)),max(p(52,:));
    min(p(53,:)),max(p(53,:));
    min(p(54,:)),max(p(54,:));
    min(p(55,:)),max(p(55,:));
    min(p(56,:)),max(p(56,:));
min(p(57,:)),max(p(57,:));
    min(p(58,:)),max(p(58,:));
    min(p(59,:)),max(p(59,:));
    min(p(60,:)),max(p(60,:));
    min(p(61,:)),max(p(61,:));
    min(p(62,:)),max(p(62,:));
    min(p(63,:)),max(p(63,:));
    min(p(64,:)),max(p(64,:));
min(p(65,:)),max(p(65,:));
    min(p(66,:)),max(p(66,:));
    min(p(67,:)),max(p(67,:));
    min(p(68,:)),max(p(68,:));
    min(p(69,:)),max(p(69,:));
    min(p(70,:)),max(p(70,:));
    min(p(71,:)),max(p(71,:));
    min(p(72,:)),max(p(72,:));
min(p(73,:)),max(p(73,:));
    min(p(74,:)),max(p(74,:));
    min(p(75,:)),max(p(75,:));
    min(p(76,:)),max(p(76,:));
    min(p(77,:)),max(p(77,:));
    min(p(78,:)),max(p(78,:));
    min(p(79,:)),max(p(79,:));
    min(p(80,:)),max(p(80,:));
min(p(81,:)),max(p(81,:));
    min(p(82,:)),max(p(82,:));
    min(p(83,:)),max(p(83,:));
    min(p(84,:)),max(p(84,:));
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        min(p(85,:)),max(p(85,:));
        min(p(86,:)),max(p(86,:));
        min(p(87,:)),max(p(87,:));
        min(p(88,:)),max(p(88,:));
min(p(89,:)),max(p(89,:));
        min(p(90,:)),max(p(90,:));
        min(p(91,:)),max(p(91,:));
        min(p(92,:)),max(p(92,:));
        min(p(93,:)),max(p(93,:));
        min(p(94,:)),max(p(94,:));
        min(p(95,:)),max(p(95,:));
        min(p(96,:)),max(p(96,:));
min(p(97,:)),max(p(97,:));
        min(p(98,:)),max(p(98,:));
        min(p(99,:)),max(p(99,:));
        min(p(100,:)),max(p(100,:));
        min(p(101,:)),max(p(101,:));
        min(p(102,:)),max(p(102,:));
        min(p(103,:)),max(p(103,:));
        min(p(104,:)),max(p(104,:));
min(p(105,:)),max(p(105,:));
        min(p(106,:)),max(p(106,:));
        min(p(107,:)),max(p(107,:));
        min(p(108,:)),max(p(108,:));
        min(p(109,:)),max(p(109,:));
        min(p(110,:)),max(p(110,:));
        min(p(111,:)),max(p(111,:));
        min(p(112,:)),max(p(112,:));
min(p(113,:)),max(p(113,:));
        min(p(114,:)),max(p(114,:))
];

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```

net=newff(Pr,[6,1],{'tansig','purelin'},'trainlm');
net.trainParam.epochs=500;
net.trainParam.goal=0.001;
net.trainParam.show=1;
net.trainParam.lr=0.05;
net =train(net,p,T);
U=U';

```